Polynomial Rectification of the Shastri Trefoil and a Harmonic Figure-Eight

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Abstract

We give an explicit implementation of coordinate transfer through affine modifications, including repeated exceptional factors. We then construct polynomial automorphisms rectifying the Shastri trefoil and a harmonic figure-eight over $\mathbb{Q}(\sqrt{-3})$, with explicit polynomial inverses.

The results were obtained on 24 October 2026 through interactive work with GPT Astra. The main polynomial identities and both explicit rectifications have been checked by exact symbolic computation. A reproducible computational certificate is provided in `verify.py`, with its run log in `verification.log`.

1 Methods

1.1 The embedding problem and its sources

The embedding problem asks whether every closed embedding $\gamma : \mathbb{A}^1_{\mathbb{C}} \hookrightarrow \mathbb{A}^3_{\mathbb{C}}$ admits $\Phi \in \mathrm{Aut}_{\mathbb{C}}(\mathbb{A}^3)$ with $\Phi(\gamma(t)) = (t, 0, 0)$. The plane case is the theorem of Abhyankar–Moh and Suzuki [1, 2]. The spatial question has a different history: Abhyankar's Kyoto 1977 conjectures are recorded in [3, Chapter 11, pp. 413–414], as documented in [4, Introduction, pp. 115–116] and [5, §1, p. 11]. In particular, the historical conjecture asserted the existence of nonrectifiable spatial lines.

Shastri's trefoil is taken from [5, §3, Example 1]. The harmonic figure-eight comes from the Chebyshev-knot literature [6, Introduction, Definition 1.2 and the following paragraph]. The coordinate-lift strategy has a precedent in [4, Proposition 3, pp. 117–118]: starting with a polynomial restricting to t, one changes it by an element of the curve ideal. Our output consists of explicit automorphisms, inverses and certificates for the two specified parametrizations.

1.2 Exact division and global polynomial identities

Work over a field k of characteristic zero. A polynomial map is called an automorphism when a polynomial inverse is supplied. The basic transition is

$$\boxed{\text{invertible after localization} \;+\; \text{polynomial in both directions} \;\Longrightarrow\; \text{polynomial automorphism.}}$$

Indeed, for $0 \neq m \in k[x]$, the inclusion

$$k[x, y, z] \hookrightarrow k[x, 1/m, y, z]$$

is injective. Once both maps have polynomial components, composition identities proved after adjoining $1/m$ hold in the original polynomial ring. Thus the fibres $m(x) = 0$ are included.

1.3 Transfer of coordinates

Suppose polynomial inverses are given:

$$\begin{aligned} A(x,y,z) &= (x, H(x,y,z), W(x,y,z)),\\ A^{-1}(x,T,V) &= (x, R(x,T,V), S(x,T,V)). \end{aligned} \tag{1}$$

Choose $0 \neq m(x) \in k[x]$ and $p(x,y) \in k[x,y]$. The modification

$$\beta_m(x,y,z) = (x, p(x,y) + m(x)z, y)$$

is invertible after adjoining $1/m$. We seek a correction that makes the complete transfer and its inverse polynomial.

Theorem 1.1 (Sufficient transfer criterion)**.** *Assume* $\deg m > 0$ *and put* $B = k[x]/(m)$*. Write*

$$c(y) = H(x, p(x,y), y) = c_0 + c_1 y + \sum_{j=2}^{d} c_j y^j \quad \text{in } B[y].$$

If c_1 *is a unit in* B *and every* c_j*,* $j \geq 2$*, is nilpotent, then* $H(x, p(x,y) + m(x)z, y)$ *extends to a polynomial automorphism preserving* x*. Both the extension and its inverse are obtained by finite polynomial operations.*

The existence conclusion is a special case of the residual-variable theorem of Bhatwadekar–Dutta [10]; the constructive relation to affine modifications is explained below. We give the formulas and an elementary proof in order to make the companions and inverses explicit.

Over an algebraic closure, the hypothesis is equivalent to

$$\deg_y H(r, p(r,y), y) = 1 \qquad \text{for every root } r \text{ of } m.$$

The construction below retains the full multiplicities in m.

Step 1: invert the centre modulo m**.** Let $J = (c_2, \dots, c_d) \subset B$ and choose $e \geq 1$ with $J^e = 0$. One may take the largest exponent in the irreducible factorization of m. In $B[T]$, set

$$\begin{aligned} \psi_0(T) &= c_1^{-1}(T - c_0),\\ \psi_{j+1}(T) &= c_1^{-1}\left(T - c_0 - \sum_{\ell=2}^{d} c_\ell \psi_j(T)^\ell\right), \qquad 0 \leq j < e. \end{aligned} \tag{2}$$

The differences satisfy $\psi_{j+1} - \psi_j \in J^{j+1}[T]$: subtract successive iterations and factor every difference of powers. Consequently the iteration stabilizes, and

$$c(\psi_e(T)) = T, \qquad \psi_e(c(y)) = y. \tag{3}$$

For the second identity, $c(v) - c(y)$ is $v - y$ times $c_1 + n$, with $n \in JB[y,v]$. After substituting $v = \psi_e(c(y))$, this factor is a unit; its inverse is a finite geometric sum. Cancel it to obtain the identity.

Step 2: choose the correction. Take any representative $\varphi(x,T) \in k[x,T]$ of

$$\varphi(x,T) \equiv W(x, p(x, \psi_e(T)), \psi_e(T)) \pmod{m(x)}. \tag{4}$$

For example, choose the remainder of degree less than $\deg m$ in x.

Step 3: form the maps. Define $\mathcal{A}$ componentwise by

$$\begin{aligned}\mathcal{A}_1(x,y,z) &= x,\\ \mathcal{A}_2(x,y,z) &= H(x,p(x,y)+m(x)z,y),\\ &\qquad W(x,p(x,y)+m(x)z,y)\\ \mathcal{A}_3(x,y,z) &= \frac{-\varphi(x,H(x,p(x,y)+m(x)z,y))}{m(x)}.\end{aligned} \tag{5}$$

Its inverse is

$$\begin{aligned}(\mathcal{A}^{-1})_1(x,T,Q) &= x,\\ (\mathcal{A}^{-1})_2(x,T,Q) &= S(x,T,m(x)Q+\varphi(x,T)),\\ &\qquad R(x,T,m(x)Q+\varphi(x,T))\\ (\mathcal{A}^{-1})_3(x,T,Q) &= \frac{-p\big(x,S(x,T,m(x)Q+\varphi(x,T))\big)}{m(x)}.\end{aligned} \tag{6}$$

Every displayed quotient is exact polynomial division of the entire numerator.

Step 4: prove both divisions exact. Equations (3)–(4) give

$$\varphi(x,H(x,p(x,y),y)) \equiv W(x,p(x,y),y) \pmod{m}.$$

Replacing p by $p+mz$ preserves this congruence, proving the divisibility in (5).

Apply A^{-1} to the two centre identities. In $B[T]$ this gives

$$\begin{aligned}R(x,T,\varphi(x,T)) &= p(x,\psi_e(T)),\\ S(x,T,\varphi(x,T)) &= \psi_e(T).\end{aligned}$$

Replacing φ by $mQ+\varphi$ preserves the congruences. Their difference proves divisibility of the numerator in (6) by the full m.

Step 5: conclude global invertibility. Over $k[x,1/m]$, let

$$C_m(x,T,V) = \left(x,T,\frac{V-\varphi(x,T)}{m(x)}\right), \qquad C_m^{-1}(x,T,Q) = (x,T,m(x)Q+\varphi(x,T)).$$

Then

$$\mathcal{A} = C_m \circ A \circ \beta_m, \qquad \mathcal{A}^{-1} = \beta_m^{-1} \circ A^{-1} \circ C_m^{-1}.$$

Both compositions are identities after localization. Polynomiality and injectivity now give

$$\mathcal{A}^{-1} \circ \mathcal{A} = \mathrm{id}, \qquad \mathcal{A} \circ \mathcal{A}^{-1} = \mathrm{id}$$

globally. This proves Theorem 1.1. If m is a nonzero constant, take $\varphi = 0$; both divisions are by a scalar and the same formulas apply. Changing φ to $\varphi + mD(x,T)$ only changes the third coordinate by the polynomial shear $-D(x,\mathcal{A}_2)$.

1.4 Relation to established constructions

The map β_m is the affine modification with divisor $m=0$ and centre ideal $(m, Y-p(x,Z))$. Under $A=(x,H,W)$, inversion of $H(x,p(x,Z),Z)$ modulo the full m identifies this centre with $(m,V-\varphi(x,T))$. Thus the transferred map is the lift between these two modifications furnished by [7, Proposition 2.1]; [7, Corollary 2.2] gives the centre-preserving case.

The quotient-ring step uses the classical description of one-variable automorphisms over a ring: a unit linear coefficient and nilpotent higher coefficients [8, Proposition 1]. Related finite constructions and transfer arguments occur in [8, Theorem 4 and §5]. A particularly close formula is recalled in [9, §1, Lemma 1.1]: if $b \circ c = c \circ b = \mathrm{id} \pmod m$, then

$$F(u,v) = \left(mv + c(u), \frac{b(mv + c(u)) - u}{m}\right)$$

is an automorphism, with inverse

$$F^{-1}(U,V) = \left(b(U) - mV, \frac{U - c(b(U) - mV)}{m}\right).$$

Both divisions are exact, and cancellation proves the inverse identities.

There is also an existence proof using [10]. Over the base $k[x]$, $H \circ \beta_m$ is a coordinate on the generic fibre and on every fibre where m is nonzero. On a fibre dividing m, the hypothesis makes it linear with nonzero slope in y. It is therefore a residual coordinate in two variables, so the theorem of Bhatwadekar–Dutta applies. The proof above supplies explicit formulas without invoking that theorem. No novelty claim is made for the general lifting principle; the concrete rectifiers and their certificates are the focus here.

1.5 From a relative coordinate to a straight line

Suppose (1) holds and $\gamma(t) = (b(t), c(t), d(t))$ satisfies $H(\gamma(t)) = t$. Put $w(t) = W(\gamma(t))$. The rectifier is

$$\Phi(x,y,z) = \big(H(x,y,z),\ x - b(H(x,y,z)),\ W(x,y,z) - w(H(x,y,z))\big). \tag{7}$$

Its inverse is explicitly

$$\begin{aligned}
(\Phi^{-1})_1(X,Y,Z) &= Y + b(X),\\
(\Phi^{-1})_2(X,Y,Z) &= R(Y + b(X), X, Z + w(X)),\\
(\Phi^{-1})_3(X,Y,Z) &= S(Y + b(X), X, Z + w(X)).
\end{aligned} \tag{8}$$

These maps compose A with a permutation and two polynomial shears. Thus $\Phi(\gamma(t)) = (t,0,0)$. The input is an already verified coordinate pair (H,W), not merely a polynomial lifting the parameter.

2 The Shastri trefoil

We work over

$$K = \mathbb{Q}[a]/(a^2 - a + 1) \simeq \mathbb{Q}(\sqrt{-3}), \qquad a = \frac{1 + i\sqrt{3}}{2}, \qquad a(1-a) = 1.$$

The input curve, given in [5, §3, Example 1, pp. 14–15], is

$$\gamma_T(t) = (t^3 - 3t,\ t^4 - 4t^2,\ t^5 - 10t). \tag{9}$$

Shastri already gives the parameter lift $yz - x^3 - 5xy + 2z - 7x$; the construction below supplies a coordinate lift together with a complete automorphism and its inverse.

2.1 The base coordinate pair and its inverse

Define

$$\begin{aligned} H(x,y,z) = {} & ax^3y^2 + 2x^2yz + (1-a)xz^2 \\ & + \big((a-1)x^3 - ax\big)y + (-1 + ax^2 - ax^4)z \\ & - 2x^5 + (6-2a)x^3 - 2(1-a)x, \end{aligned} \tag{10}$$

and

$$\begin{aligned} f(x,T) = {} & -T + x\big((1-a)T^2 + 2a - 2\big) \\ & + x^2(2aT^3 - 5aT) + x^3(-5T^4 + 15T^2 - 2a) \\ & + x^4\big(14(1-a)T^5 + 50(a-1)T^3 + (29 - 24a)T\big). \end{aligned} \tag{11}$$

The two exact quotients

$$\begin{aligned} W(x,y,z) &= \frac{a\big(axy + z - f(x, H(x,y,z))\big)}{x^5}, \\ \chi(x,T) &= \frac{a\big(T - H(x,0,f(x,T))\big)}{x^5} \end{aligned} \tag{12}$$

are polynomials. Reduction by $a^2 - a + 1$ and coefficientwise division by x^5 give their full coefficients in the certificate. Define

$$\begin{aligned} R(x,T,V) = {} & -\chi(x,T) \\ & + (-1 + ax^2 - ax^4 + 2(1-a)xf(x,T))V \\ & + (1-a)^2x^6V^2, \\ S(x,T,V) = {} & f(x,T) + (1-a)x^5V - axR(x,T,V). \end{aligned} \tag{13}$$

Then

$$A_0 = (x, H, W), \qquad A_0^{-1} = (x, R, S).$$

2.2 Certificate of both inverse identities

The exact polynomial checks are

$$x^5W(x,y,z) = a\big(axy + z - f(x, H(x,y,z))\big), \tag{14}$$

$$H(x, R(x,T,V), S(x,T,V)) = T, \tag{15}$$

$$axR(x,T,V) + S(x,T,V) - f(x,T) = (1-a)x^5V, \tag{16}$$

$$R(x, H(x,y,z), W(x,y,z)) = y, \tag{17}$$

$$S(x, H(x,y,z), W(x,y,z)) = z. \tag{18}$$

Substitute R, S into (14) and use (15)–(16):

$$x^5 W(x, R, S) = a(1-a)x^5 V = x^5 V.$$

Cancellation in $K[x, T, V]$ gives $W(x, R, S) = V$. Together with (15), this proves $A_0 \circ A_0^{-1} = \mathrm{id}$; (17)–(18) prove the opposite composition. The divisions in (12) and this cancellation are polynomial operations, valid also on the fibre $x = 0$.

2.3 Normalization and final rectifier

The affine maps

$$M(x, y, z) = (x, y + 2, z - 5x), \qquad M^{-1}(x, y, z) = (x, y - 2, z + 5x)$$

give

$$M(\gamma_T(t)) = (t^3 - 3t,\ t^4 - 4t^2 + 2,\ t^5 - 5t^3 + 5t).$$

Put

$$h_T(x, y, z) = H(x, y + 2, z - 5x), \qquad w_T(x, y, z) = W(x, y + 2, z - 5x).$$

Exact substitution yields

$$h_T(\gamma_T(t)) = t, \qquad w_T(\gamma_T(t)) = 8 - 14t^2.$$

The final components are therefore

$$\begin{aligned} (\Phi_T)_1(x, y, z) &= h_T(x, y, z), \\ (\Phi_T)_2(x, y, z) &= x - h_T(x, y, z)^3 + 3h_T(x, y, z), \\ (\Phi_T)_3(x, y, z) &= w_T(x, y, z) - 8 + 14h_T(x, y, z)^2. \end{aligned} \tag{19}$$

The inverse, on input (X, Y, Z), is

$$\begin{aligned} (\Phi_T^{-1})_1(X, Y, Z) &= X^3 - 3X + Y, \\ (\Phi_T^{-1})_2(X, Y, Z) &= R(X^3 - 3X + Y, X, Z + 8 - 14X^2) - 2, \\ (\Phi_T^{-1})_3(X, Y, Z) &= S(X^3 - 3X + Y, X, Z + 8 - 14X^2) \\ &\quad + 5(X^3 - 3X + Y). \end{aligned} \tag{20}$$

Explicitly, with

$$L_T(x, T, V) = (T, x - T^3 + 3T, V - 8 + 14T^2),$$

the transition is

$$\boxed{\Phi_T = L_T \circ A_0 \circ M, \qquad \Phi_T^{-1} = M^{-1} \circ A_0^{-1} \circ L_T^{-1}.}$$

Each factor is a global polynomial automorphism, and

$$\boxed{\Phi_T(\gamma_T(t)) = (t, 0, 0), \qquad \deg \Phi_T = (5, 15, 24), \qquad \det D\Phi_T = 1.} \tag{21}$$

The degree triple records the displayed construction.

2.4 Coordinate-only formulas

The formulas below use only the source coordinates (x, y, z), the target coordinates (X, Y, Z), and the fixed scalar a from the preceding construction.

Forward map

Component 1.

$$\begin{aligned}(\Phi_T)_1(x,y,z) = \; & -z + x\big((1-a)z^2 - ay + 3\big) \\ & + x^2 z(2y + 11a - 6) \\ & + x^3\big(ay^2 + (5a-11)y + 9 - 26a\big) \\ & - ax^4 z + (5a-2)x^5\end{aligned}$$

Component 2.

$$\begin{aligned}(\Phi_T)_2(x,y,z) = x - \Big[& -z + x\big((1-a)z^2 - ay + 3\big) \\ & + x^2 z(2y + 11a - 6) \\ & + x^3\big(ay^2 + (5a-11)y + 9 - 26a\big) \\ & - ax^4 z + (5a-2)x^5\Big]^3 \\ + 3\Big[& -z + x\big((1-a)z^2 - ay + 3\big) \\ & + x^2 z(2y + 11a - 6) \\ & + x^3\big(ay^2 + (5a-11)y + 9 - 26a\big) \\ & - ax^4 z + (5a-2)x^5\Big]\end{aligned}$$

Component 3.

$$\begin{aligned}(\Phi_T)_3(x,y,z) = \; & -y + z^6(42a-42) + z^4(175-175a) + z^2(147a-148) \\ & + x\Big[-252yz^5 + 700yz^3 - 296yz - 120az^7 + z^5(756-686a) + z^3(2594a-2148) \\ & \quad + z(954-1063a)\Big] \\ & + x^2\Big[-630ay^2z^4 + 1050ay^2z^2 + y^2(-148a-1) + yz^6(840-840a) \\ & \quad + yz^4(350a+3430) + yz^2(1342a-7782) + y(1069-120a) + 135z^8 \\ & \quad + z^6(2520a+1050) + z^4(4646a-7778) + z^2(10277-14168a) + 1910a - 1536\Big] \\ & + x^3\Big[y^3z^3(840-840a) + y^3z(700a-700) + 2520y^2z^5 + y^2z^3(7560a-700) \\ & \quad + y^2z(-6436a-1346) + 1080ayz^7 + yz^5(21420a-15120) + yz^3(-12528a-18576) \\ & \quad + yz(28316-7806a) + z^9(70a-70) + z^7(2000a-5240) \\ & \quad + z^5(-21414a-20590) + z^3(52712-20252a) + z(33927a-22907)\Big] \\ & + x^4\Big[630y^4z^2 - 175y^4 + 4200ay^3z^4 + y^3z^2(6860a-7560) + y^3(2144-2594a) \\ & \quad + y^2z^6(3780a-3780) + y^2z^4(15750a-53550) + y^2z^2(18792-46644a) \\ & \quad + y^2(10233a+3915) - 630yz^8 + yz^6(-22680a-14000) + yz^4(107086-210036a) \\ & \quad + yz^2(97440a+60624) + y(11032a-33842) - 14az^{10} + z^8(1890-3290a) \\ & \quad + z^6(76600-52550a) + z^4(99328a+168972) + z^2(68747a-210957) - 30180a - 20159\Big]\end{aligned}$$

$$
\begin{aligned}
&+ x^5\Big[252ay^5z + y^4z^3\,(4200a-4200) + y^4z\,(-350a-3430) - 7560y^3z^5\\
&\quad + y^3z^3\,(-50400a-21000) + y^3z\,(31088-18560a) - 2520ay^2z^7\\
&\quad + y^2z^5\,(68040-110040a) + y^2z^3\,(420104-205900a) + y^2z\,(157992a-97500)\\
&\quad + yz^9\,(140-140a) + yz^7\,(26320-11200a) + yz^5\,(144520a+315300)\\
&\quad + yz^3\,(1072852a-397504) + yz\,(-284634a-136916) + z^9\,(420a+350)\\
&\quad + z^7\,(56440a-11780) + z^5\,(575490a-576910) + z^3\,(-290692a-749608)\\
&\quad + z\,(438422-151320a)\Big]\\
&+ x^6\Big[y^6\,(42a-42) - 2520y^5z^2 + y^5\,(70-756a) - 9450ay^4z^4\\
&\quad + y^4z^2\,(37800-53550a) + y^4\,(3132a+4638) + y^3z^6\,(5880-5880a)\\
&\quad + y^3z^4\,(183400-70000a) + y^3z^2\,(214236a+205900) + y^3\,(20120a-52640)\\
&\quad + 630y^2z^8 + y^2z^6\,(52920a+39200) + y^2z^4\,(1150100a-361850)\\
&\quad + y^2z^2\,(1012212a-1608756) + y^2\,(142424-210592a) + yz^8\,(6930a-3780)\\
&\quad + yz^6\,(312620a-395480) + yz^4\,(-10400a-2878100) + yz^2\,(874278-3116768a)\\
&\quad + y\,(287148a+150558) + z^8\,(-1890a-13090) + z^6\,(-476020a-114340)\\
&\quad + z^4\,(2482481-3435465a) + z^2\,(586308a+1916345) + 141719a - 362989\Big]\\
&+ x^7\Big[-840ay^6z + y^5z^3\,(7560-7560a) + y^5z\,(21420-6300a) + 8820y^4z^5\\
&\quad + y^4z^3\,(113400a+70000) + y^4z\,(210084a-107134) + 1680ay^3z^7\\
&\quad + y^3z^5\,(184240a-105840) + y^3z^3\,(1051000a-1534200) + y^3z\,(-397888a-674268)\\
&\quad + y^2z^7\,(12600a-27720) + y^2z^5\,(-249780a-937860) + y^2z^3\,(27400-5784900a)\\
&\quad + y^2z\,(3112642-2236156a) + yz^7\,(15400-120120a) + yz^5\,(2867220-3546060a)\\
&\quad + yz^3\,(13746980-3789088a) + yz\,(4991608a-1182968) - 70z^9 + z^7\,(199920-66220a)\\
&\quad + z^5\,(2066230a+2183680) + z^3\,(12151256a-6212842) + z\,(-785906a-2703828)\Big]\\
&+ x^8\Big[y^7\,(120-120a) + 3780y^6z^2 + y^6\,(2520a+1050) + 8820ay^5z^4\\
&\quad + y^5z^2\,(110040a-68040) + y^5\,(20590a-42020) + y^4z^6\,(2940a-2940)\\
&\quad + y^4z^4\,(98000a-230300) + y^4z^2\,(-362950a-788250) + y^4\,(99520-267952a)\\
&\quad + y^3z^6\,(-35280a-29400) + y^3z^4\,(418300-1981400a)\\
&\quad + y^3z^2\,(5792800-5758800a) + y^3\,(292900a+743274) + y^2z^6\,(421400-366520a)\\
&\quad + y^2z^4\,(8874900-1679100a) + y^2z^2\,(14978766a+5649384) + y^2\,(1892310a-2488988)\\
&\quad - 560ayz^8 + yz^6\,(945980a+457660) + yz^4\,(21318610a-10445050)\\
&\quad + yz^2\,(17722764a-36463092) + y\,(801004-3475076a) + z^8\,(3290-1260a)\\
&\quad + z^6\,(1161050a-1627570) + z^4\,(-3444640a-14775140) + z^2\,(8544658-25600678a)\\
&\quad + 513084a + 1655387\Big]\\
&+ x^9\Big[1080ay^7z + y^6z^3\,(5880a-5880) + y^6z\,(14000a-36680) - 3528y^5z^5\\
&\quad + y^5z^3\,(-105840a-78400) + y^5z\,(145400-460700a) + y^4z^5\,(52920-97020a)\\
&\quad + y^4z^3\,(1983400-1563100a) + y^4z\,(20300a+2880050) + y^3z^5\,(111720a+733040)\\
&\quad + y^3z^3\,(9631400a+2214800) + y^3z\,(13757220a-10013780) + y^2z^7\,(1960-1960a)\\
&\quad + y^2z^5\,(4223520a-2868180) + y^2z^3\,(21657360a-42775340)\\
&\quad + y^2z\,(-18841866a-17630430) + yz^7\,(14560a+8680) + yz^5\,(-2928540a-6935160)\\
&\quad + yz^3\,(14365760-73365260a) + yz\,(51189024-33903048a) + z^7\,(41300a-59220)\\
&\quad + z^5\,(7559050-8869506a) + z^3\,(54487632-7161702a) + z\,(29896586a-5286324)\Big]\\
&+ x^{10}\Big[y^8\,(135a-135) - 2520y^7z^2 + y^7\,(-3240a-2000) - 2940ay^6z^4
\end{aligned}
$$

$$
\begin{aligned}
&+ y^6 z^2 (52920 - 92120a) + y^6 (76820 - 52550a) + y^5 z^4 (97020 - 44100a) \\
&+ y^5 z^2 (253380a + 937860) + y^5 (580860a - 4720) + y^4 z^4 (1058400a - 142100) \\
&+ y^4 z^2 (8894400a - 7251300) + y^4 (930160a - 3440585) + 3920 y^3 z^6 \\
&+ y^3 z^4 (2229500a - 7060200) + y^3 z^2 (-21345940a - 21567520) \\
&+ y^3 (6314380 - 12160460a) + y^2 z^6 (70560a - 45080) + y^2 z^4 (7645290 - 24904950a) \\
&+ y^2 z^2 (110776650 - 88346940a) + y^2 (8740740a + 16847244) + y z^6 (-119140a - 248640) \\
&+ y z^4 (44462990 - 5745780a) + y z^2 (143679990a + 19834860) + y (24411924a - 29861264) \\
&+ z^8 (140a - 140) + z^6 (556220 - 576660a) + z^4 (38669030a - 18846150) \\
&+ z^2 (44597758a - 116164920) - 15005311a - 544778\Big] \\
+ x^{11}\Big[&-630 a y^8 z + y^7 z^3 (1680 - 1680a) + y^7 z (26320 - 11200a) \\
&+ y^6 z^3 (35280a + 29400) + y^6 z (397480a - 84860) + y^5 z^3 (733040a - 848680) \\
&+ y^5 z (650040a - 3561660) + 4900 a y^4 z^5 + y^4 z^3 (-4881100a - 2200100) \\
&+ y^4 z (10786990 - 21525790a) + y^3 z^5 (41160a - 119560) \\
&+ y^3 z^3 (33534200 - 22908040a) + y^3 z (15540700a + 58795940) \\
&+ y^2 z^5 (338100 - 962640a) + y^2 z^3 (79280140a + 9872080) \\
&+ y^2 z (163563900a - 145378620) - 840 y z^7 + y z^5 (20720a + 2994600) \\
&+ y z^3 (76662620a - 156536960) + y z (-146104300a - 86754860) + z^7 (3920 - 5460a) \\
&+ z^5 (4513530a - 2993810) + z^3 (14467230 - 101150360a) + z (135041266 - 79575830a)\Big] \\
+ x^{12}\Big[&y^9 (70 - 70a) + 630 y^8 z^2 + y^8 (1890a + 1400) + y^7 z^2 (27720a - 15120) \\
&+ y^7 (44660a - 56840) + y^6 z^2 (-58800a - 366520) + y^6 (-487120a - 107140) \\
&+ y^5 z^4 (3920a - 3920) + y^5 z^2 (2958900 - 4261320a) + y^5 (4318970 - 2138760a) \\
&+ y^4 z^4 (-83300a - 39200) + y^4 z^2 (8294340a + 17102010) + y^4 (18705470a - 4032110) \\
&+ y^3 z^4 (1369200 - 838600a) + y^3 z^2 (89373780a - 81127660) + y^3 (5512120a - 54536680) \\
&- 2100 a y^2 z^6 + y^2 z^4 (6722800a - 403900) + y^2 z^2 (-126562290a - 111012540) \\
&+ y^2 (74530190 - 116689560a) + y z^6 (28140 - 7140a) + y z^4 (5749280a - 19792780) \\
&+ y z^2 (311292480 - 255832080a) + y (57536360a + 78070540) + z^6 (87780a - 44660) \\
&+ z^4 (9049550 - 21721090a) + z^2 (153817680a + 43554560) + 51428936a - 66560746\Big] \\
+ x^{13}\Big[&140 a y^9 z + y^8 z (3150a - 6930) + y^7 z (17080 - 121800a) - 1960 y^6 z^3 \\
&+ y^6 z (1424640 - 428260a) + y^5 z^3 (54880 - 76440a) + y^5 z (10256820a - 3447780) \\
&+ y^4 z^3 (497000a + 636300) + y^4 z (3306840a - 44795330) + y^3 z^5 (2800 - 2800a) \\
&+ y^3 z^3 (6862800a - 7862400) + y^3 z (88886160 - 160214920a) + y^2 z^5 (46200a + 12600) \\
&+ y^2 z^3 (-26146380a - 7826860) + y^2 z (67566960a + 251519220) + y z^5 (170520a - 388920) \\
&+ y z^3 (77719460 - 40983200a) + y z (392466420a - 322381340) + 140 a z^7 \\
&+ z^5 (258860 - 786240a) + z^3 (65853920a - 11843680) + z (-119811310a - 116823990)\Big] \\
+ x^{14}\Big[&y^{10} (14a - 14) + y^9 (-420a - 350) + y^8 (15260 - 13090a) - 560 a y^7 z^2 \\
&+ y^7 (143780a + 60340) + y^6 z^2 (27440 - 5880a) + y^6 (1129520a - 1725850) \\
&+ y^5 z^2 (538020a - 277620) + y^5 (-8482810a - 497476) + 2100 y^4 z^4 \\
&+ y^4 z^2 (-1089200a - 3971100) + y^4 (40508020 - 17155960a) + y^3 z^4 (63000a - 53200) \\
&+ y^3 z^2 (24116120 - 28298340a) + y^3 (108944120a - 26573740) \\
&+ y^2 z^4 (-429660a - 248640) + y^2 z^2 (55559100a + 46926900) \\
&+ y^2 (26484020a - 195006220) + y z^6 (560a - 560) + y z^4 (2954840 - 1747340a) \\
&+ y z^2 (135349780a - 181368660) + y (130474810 - 237389530a) + z^6 (-3780a - 560)
\end{aligned}
$$

$$
\begin{aligned}
&\quad + z^4\left(4358195a - 708675\right) + z^2\left(-122780160a - 9142480\right) + 32610896a + 85823498\Big] \\
&+ x^{15}\Big[y^8 z\left(70 - 70a\right) + y^7 z\left(4480a + 280\right) + y^6 z\left(46340a - 131880\right) + 840a y^5 z^3 \\
&\quad + y^5 z\left(568960 - 1685320a\right) + y^4 z^3\left(2100a - 35700\right) + y^4 z\left(11280430 - 282380a\right) \\
&\quad + y^3 z^3\left(428120 - 579880a\right) + y^3 z\left(58698080a - 37107000\right) - 840 y^2 z^5 \\
&\quad + y^2 z^3\left(2153340a + 2046520\right) + y^2 z\left(-55844860a - 108489920\right) + y z^5\left(13440 - 14280a\right) \\
&\quad + y z^3\left(9787120a - 13489000\right) + y z\left(232809100 - 225909200a\right) + z^5\left(46410a + 12460\right) \\
&\quad + z^3\left(38260 - 15464760a\right) + z\left(128584690a + 47607820\right)\Big] \\
&+ x^{16}\Big[y^8\left(210a - 350\right) + y^7\left(-11200a - 840\right) + y^6 z^2\left(140a - 140\right) \\
&\quad + y^6\left(223720 - 107100a\right) + y^5 z^2\left(1260 - 10920a\right) + y^5\left(1978620a - 302540\right) \\
&\quad + y^4 z^2\left(242340 - 21840a\right) + y^4\left(2975840a - 12210450\right) - 560a y^3 z^4 \\
&\quad + y^3 z^2\left(2736860a - 1806140\right) + y^3\left(21668920 - 48820380a\right) + y^2 z^4\left(840a + 16800\right) \\
&\quad + y^2 z^2\left(-6132000a - 8696520\right) + y^2\left(15559380a + 93710320\right) + y z^4\left(158130a - 144130\right) \\
&\quad + y z^2\left(37263660 - 31518060a\right) + y\left(154112960a - 126732800\right) + 70 z^6 \\
&\quad + z^4\left(-333830a - 118860\right) + z^2\left(34578030a + 5182450\right) - 57839390a + 42459540\Big] \\
&+ x^{17}\Big[y^6 z\left(1400 - 840a\right) + y^5 z\left(51240a - 10920\right) + y^4 z^3\left(140 - 140a\right) \\
&\quad + y^4 z\left(74340a - 762300\right) + y^3 z^3\left(10080a - 1960\right) + y^3 z\left(4000640 - 6639360a\right) \\
&\quad + y^2 z^3\left(-16100a - 141680\right) + y^2 z\left(9412620a + 18990720\right) + 140a y z^5 \\
&\quad + y z^3\left(871640 - 959000a\right) + y z\left(55149080a - 57934520\right) + z^5\left(-210a - 1610\right) \\
&\quad + z^3\left(1521520a + 608440\right) + z\left(-44786060a - 12873480\right)\Big] \\
&+ x^{18}\Big[y^6\left(700a - 2940\right) + y^5\left(29820 - 86520a\right) + y^4 z^2\left(1260a - 2100\right) \\
&\quad + y^4\left(941640 - 79800a\right) + y^3 z^2\left(24360 - 68880a\right) + y^3\left(6626340a - 3712940\right) \\
&\quad + y^2 z^4\left(70a - 70\right) + y^2 z^2\left(128940a + 612360\right) + y^2\left(-6090000a - 16976680\right) \\
&\quad + y z^4\left(770 - 3220a\right) + y z^2\left(3336060a - 3107580\right) + y\left(39160240 - 40771780a\right) \\
&\quad + z^4\left(4690a + 15960\right) + z^2\left(-4383120a - 1712200\right) + 25782295a + 10347625\Big] \\
&+ x^{19}\Big[y^4 z\left(8820 - 2100a\right) + y^3 z\left(213920a - 110040\right) + y^2 z^3\left(1400 - 840a\right) \\
&\quad + y^2 z\left(-469980a - 1331400\right) + y z^3\left(28840a - 12600\right) + y z\left(6142360 - 6261360a\right) \\
&\quad + z^5\left(14 - 14a\right) + z^3\left(-45780a - 83300\right) + z\left(7302330a + 2428860\right)\Big] \\
&+ x^{20}\Big[y^4\left(-2380a - 10220\right) + y^3\left(175000 - 248640a\right) + y^2 z^2\left(2100a - 8820\right) \\
&\quad + y^2\left(648620a + 1147160\right) + y z^2\left(80220 - 127400a\right) + y\left(4905810a - 5199810\right) \\
&\quad + z^4\left(210a - 350\right) + z^2\left(228900a + 226310\right) - 5332390a + 1292060\Big] \\
&+ x^{21}\Big[y^2 z\left(4760a + 20440\right) + y z\left(274540a - 233240\right) + z^3\left(2940 - 700a\right) \\
&\quad + z\left(-563290a - 272370\right)\Big] \\
&+ x^{22}\Big[y^2\left(-16170a - 12950\right) + y\left(255570 - 223300a\right) + z^2\left(-2380a - 10220\right) + 536970a + 74620\Big] \\
&+ x^{23}\Big[z\left(16170a + 12950\right)\Big] \\
&+ x^{24}\Big[-21350a + 1302\Big]
\end{aligned}
$$

Inverse map

Component 1.

$$(\Phi_T^{-1})_1(X,Y,Z) = X^3 - 3X + Y.$$

Component 2.

$$\begin{aligned}
(\Phi_T^{-1})_2(X,Y,Z) = {} & X^6\,(42a-42) + X^4\,(175-175a) + X^2\,(147a-162) + 8 \\
& + (X^3-3X+Y)\Big[-48X^7a + 280X^5a + X^3\,(20-466a) + X\,(165a-50)\Big] \\
& + (X^3-3X+Y)^2\Big[81X^8 - 490X^6 + X^4\,(45a+816) + X^2\,(-125a-225) + 4a - 6\Big] \\
& + (X^3-3X+Y)^3\Big[X^9\,(140a-140) + X^7\,(920-920a) + X^5\,(1726a-1846) \\
& \qquad + X^3\,(1070-670a) + X\,(-49a-91)\Big] \\
& + (X^3-3X+Y)^4\Big[-196X^{10}a + 1400X^8a + X^6\,(140-3312a) + X^4\,(2900a-500) \\
& \qquad + X^2\,(265-816a)\Big] \\
& + (Z+8-14X^2)\Big[-1\Big] \\
& + (Z+8-14X^2)(X^3-3X+Y)\Big[X\,(2a-2)\Big] \\
& + (Z+8-14X^2)(X^3-3X+Y)^2\Big[-2X^2a + 5a\Big] \\
& + (Z+8-14X^2)(X^3-3X+Y)^3\Big[4X^3 - 10X\Big] \\
& + (Z+8-14X^2)(X^3-3X+Y)^4\Big[X^4\,(10a-10) + X^2\,(30-30a) - a + 4\Big] \\
& + (Z+8-14X^2)(X^3-3X+Y)^5\Big[-28X^5a + 100X^3a + X\,(10-58a)\Big] \\
& + (Z+8-14X^2)^2(X^3-3X+Y)^6\Big[-a\Big]
\end{aligned}$$

Component 3.

$$
\begin{aligned}
(\Phi_T^{-1})_3(X,Y,Z) = & -X \\
& + (X^3-3X+Y)\Big[42X^6-175X^4+X^2(14a+148)-8a-3\Big] \\
& + (X^3-3X+Y)^2\Big[X^7(48a-48)+X^5(280-280a)+X^3(448a-466) \\
& \quad + X(165-120a)\Big] \\
& + (X^3-3X+Y)^3\Big[-81X^8a+490X^6a+X^4(40-861a)+X^2(350a-110)+4\Big] \\
& + (X^3-3X+Y)^4\Big[140X^9-920X^7+X^5(106a+1740)+X^3(-350a-720) \\
& \quad + X(116a-20)\Big] \\
& + (X^3-3X+Y)^5\Big[X^{10}(196a-196)+X^8(1400-1400a)+X^6(3172a-3312) \\
& \quad + X^4(2900-2400a)+X^2(551a-816)\Big] \\
& + (Z+8-14X^2)(X^3-3X+Y)\Big[a\Big] \\
& + (Z+8-14X^2)(X^3-3X+Y)^2\Big[2X\Big] \\
& + (Z+8-14X^2)(X^3-3X+Y)^3\Big[X^2(2a-2)-5a-5\Big] \\
& + (Z+8-14X^2)(X^3-3X+Y)^4\Big[-4X^3a+10Xa\Big] \\
& + (Z+8-14X^2)(X^3-3X+Y)^5\Big[10X^4-30X^2+4a\Big] \\
& + (Z+8-14X^2)(X^3-3X+Y)^6\Big[X^5(28a-28)+X^3(100-100a)+X(48a-58)\Big] \\
& + (Z+8-14X^2)^2(X^3-3X+Y)^7\Big[a-1\Big]
\end{aligned}
$$

3 The harmonic figure-eight

The second curve is

$$\gamma_H(t) = (t^3 - 3t,\ t^5 - 5t^3 + 5t,\ t^7 - 7t^5 + 14t^3 - 7t). \tag{22}$$

The coefficient in the second component is $+5t$. Indeed,

$$\gamma_H(t) = (2T_3(t/2), 2T_5(t/2), 2T_7(t/2)),$$

where $T_n(\cos\theta) = \cos(n\theta)$. This identifies the parametrization with the harmonic knot $H(3,5,7)$ of [6] by linear scalings. It is distinct from the figure-eight parametrization in [5, §3, Example 2], whose second component contains $+4t$.

3.1 Modification and correction

Take $m(x) = x$ and $p(x,y) = 2 - y^2$. The modification

$$\beta(x,y,z) = (x, xz - y^2 + 2, y)$$

satisfies

$$\boxed{\beta(\gamma_H(t)) = M(\gamma_T(t)).} \tag{23}$$

Its inverse after adjoining $1/x$ is

$$\beta^{-1}(x,Y,Z) = \left(x, Z, \frac{Y + Z^2 - 2}{x}\right).$$

The hypothesis of Theorem 1.1 follows from

$$H(0,y,z) = -z, \qquad H(0, 2 - y^2, y) = -y.$$

Thus the centre inverse is $\psi(T) = -T$, and the correction is

$$\begin{aligned} \varphi(T) &= W(0, 2 - T^2, -T) \\ &= 42(a-1)T^6 + 175(1-a)T^4 + (147a - 161)T^2 + 8. \end{aligned} \tag{24}$$

The transferred coordinates are

$$\begin{aligned} h_H(x,y,z) &= H(x, xz - y^2 + 2, y), \\ q_H(x,y,z) &= \frac{W(x, xz - y^2 + 2, y) - \varphi(h_H(x,y,z))}{x}. \end{aligned} \tag{25}$$

The inverse of (x, h_H, q_H) has components

$$\begin{aligned} R_H(x,T,Q) &= S(x, T, xQ + \varphi(T)), \\ S_H(x,T,Q) &= \frac{+R(x,T,xQ+\varphi(T))}{+S(x,T,xQ+\varphi(T))^2 - 2}{x}. \end{aligned} \tag{26}$$

Both numerators divided by x are exactly divisible. This is established by Theorem 1.1 and verified directly in the certificate.

3.2 The transition is global

Over $K[x, 1/x]$, set

$$C(x,T,V) = \left(x, T, \frac{V-\varphi(T)}{x}\right), \qquad C^{-1}(x,T,Q) = (x,T,xQ+\varphi(T)).$$

Then

$$\begin{aligned}(x, h_H, q_H) &= C \circ A_0 \circ \beta,\\ (x, R_H, S_H) &= \beta^{-1} \circ A_0^{-1} \circ C^{-1}.\end{aligned} \tag{27}$$

The individual maps β^{-1} and C require $1/x$. The complete maps in (27) are polynomial by the two exact divisions. Injectivity of localization therefore proves both composition identities in $K[x, y, z]$, including $x = 0$.

3.3 Final rectifier and inverse

Curve substitution gives

$$h_H(\gamma_H(t)) = t, \qquad q_H(\gamma_H(t)) = (1-a)(42t^3 - 49t).$$

Define

$$\begin{aligned}(\Phi_H)_1(x,y,z) &= h_H(x,y,z),\\ (\Phi_H)_2(x,y,z) &= x - h_H(x,y,z)^3 + 3h_H(x,y,z),\\ (\Phi_H)_3(x,y,z) &= q_H(x,y,z)\\ &\quad - (1-a)\big(42h_H(x,y,z)^3 - 49h_H(x,y,z)\big).\end{aligned} \tag{28}$$

The inverse is

$$\begin{aligned}(\Phi_H^{-1})_1(X,Y,Z) &= X^3 - 3X + Y,\\ (\Phi_H^{-1})_2(X,Y,Z) &= R_H(X^3 - 3X + Y, X, Z + (1-a)(42X^3 - 49X)),\\ (\Phi_H^{-1})_3(X,Y,Z) &= S_H(X^3 - 3X + Y, X, Z + (1-a)(42X^3 - 49X)).\end{aligned} \tag{29}$$

All auxiliary polynomials in these formulas have been explicitly defined in (10)–(13) and (25)–(26). The final transition is the permutation and shear

$$L_H(x,T,Q) = (T, x - T^3 + 3T, Q - (1-a)(42T^3 - 49T)),$$

so

$$\boxed{\Phi_H = L_H \circ C \circ A_0 \circ \beta.}$$

The resulting identities and degrees are

$$\boxed{\Phi_H(\gamma_H(t)) = (t,0,0), \qquad \deg \Phi_H = (7,21,41), \qquad \det D\Phi_H = -1.} \tag{30}$$

3.4 Coordinate-only formulas

The following formulas use only the input coordinates and the constant $a = (1 + i\sqrt{3})/2$. Repeated expressions are written in full.

Forward map

Component 1.

$$\begin{aligned}(\Phi_H)_1(x,y,z) = \; & -y + x\Big[(1-a)\big(ax(xz-y^2+2)+y\big)^2 \\ & + \big((a-1)x^2 - a\big)(xz-y^2+2) \\ & + ax(1-x^2)y - 2x^4 + (6-2a)x^2 - 2(1-a)\Big]\end{aligned}$$

Component 2.

$$\begin{aligned}(\Phi_H)_2(x,y,z) = x - \Big\{ & -y + x\Big[(1-a)\big(ax(xz-y^2+2)+y\big)^2 \\ & + \big((a-1)x^2 - a\big)(xz-y^2+2) \\ & + ax(1-x^2)y - 2x^4 + (6-2a)x^2 - 2(1-a)\Big]\Big\}^3 \\ + 3\Big\{ & -y + x\Big[(1-a)\big(ax(xz-y^2+2)+y\big)^2 \\ & + \big((a-1)x^2 - a\big)(xz-y^2+2) \\ & + ax(1-x^2)y - 2x^4 + (6-2a)x^2 - 2(1-a)\Big]\Big\}\end{aligned}$$

Component 3.

$$
\begin{aligned}
(\Phi_H)_3(x,y,z) = {} & y^3\left(42-42a\right)+y\left(49a-49\right)-z \\
& -120ay^7+574ay^5+y^3\left(-934a-20\right)+y\left(463a+10\right) \\
& +\left(xz-y^2+2\right)\Big[-252y^5+700y^3+y\left(-28a-296\right)\Big] \\
& +x\Big[135y^8-630y^6+y^4\left(40a+1112\right)+y^2\left(-20a-1129\right) \\
& \quad -26a-352\Big] \\
& +x(xz-y^2+2)\Big[y^6\left(840-840a\right)+y^4\left(2870a-2870\right) \\
& \quad +y^2\left(2802-2858a\right)+472a-463\Big] \\
& +x(xz-y^2+2)^2\Big[-630ay^4+1050ay^2-162a-13\Big] \\
& +x^2\Big[y^9\left(70a-70\right)+y^7\left(160-160a\right) \\
& \quad +y^5\left(830-1254a\right)+y^3\left(3784a-2444\right) \\
& \quad +y\left(711-1331a\right)\Big] \\
& +x^2(xz-y^2+2)\Big[1080ay^7-3780ay^5+y^3\left(4608a-152\right) \\
& \quad +y\left(36-2298a\right)\Big] \\
& +x^2(xz-y^2+2)^2\Big[2520y^5-5740y^3+y\left(2854-52a\right)\Big] \\
& +x^2(xz-y^2+2)^3\Big[y^3\left(840-840a\right)+y\left(700a-700\right)\Big] \\
& +x^3\Big[-14ay^{10}-140ay^8+y^6\left(2330a+580\right) \\
& \quad +y^4\left(-7550a-2400\right)+y^2\left(9147a+4135\right)-1346a+940\Big] \\
& +x^3(xz-y^2+2)\Big[-630y^8+1120y^6+y^4\left(6286-2136a\right) \\
& \quad +y^2\left(4032a-11364\right)-620a-1332\Big] \\
& +x^3(xz-y^2+2)^2\Big[y^6\left(3780a-3780\right)+y^4\left(9450-9450a\right) \\
& \quad +y^2\left(6696a-6912\right)-1133a-1149\Big] \\
& +x^3(xz-y^2+2)^3\Big[4200ay^4-5740ay^2+934a+16\Big] \\
& +x^3(xz-y^2+2)^4\Big[630y^2-175\Big] \\
& +x^4\Big[70y^9+y^7\left(160a-1420\right)+y^5\left(190a+3530\right) \\
& \quad +y^3\left(-2096a-3856\right)+y\left(10532-2220a\right)\Big] \\
& +x^4(xz-y^2+2)\Big[y^9\left(140-140a\right)+y^7\left(1120-1120a\right) \\
& \quad +y^5\left(17680a-13980\right)+y^3\left(30200-40468a\right) \\
& \quad +y\left(26882a-18320\right)\Big] \\
& +x^4(xz-y^2+2)^2\Big[-2520ay^7+3360ay^5+y^3\left(8300a+4304\right) \\
& \quad +y\left(-7332a-4044\right)\Big] \\
& +x^4(xz-y^2+2)^3\Big[-7560y^5+12600y^3+y\left(-136a-4472\right)\Big] \\
& +x^4(xz-y^2+2)^4\Big[y^3\left(4200a-4200\right)+y\left(2870-2870a\right)\Big] \\
& +x^4(xz-y^2+2)^5\Big[252ay\Big]
\end{aligned}
$$

$$
\begin{aligned}
&+ x^5\Big[140y^8 + y^6\left(1680 - 3940a\right) + y^4\left(25535a - 11419\right)\\
&\quad + y^2\left(16523 - 43464a\right) - 336a - 2356\Big]\\
&+ x^5(xz - y^2 + 2)\Big[630ay^8 + y^6\left(-8820a - 1520\right) + y^4\left(18600a + 600\right)\\
&\quad + y^2\left(4002 - 17748a\right) + 8304a + 2692\Big]\\
&+ x^5(xz - y^2 + 2)^2\Big[630y^8 + 3920y^6 + y^4\left(9800a - 44750\right)\\
&\quad + y^2\left(61704 - 16404a\right) + 4428a - 13600\Big]\\
&+ x^5(xz - y^2 + 2)^3\Big[y^6\left(5880 - 5880a\right) + y^4\left(5600a - 5600\right)\\
&\quad + y^2\left(12636a - 8300\right) - 3796a - 2444\Big]\\
&+ x^5(xz - y^2 + 2)^4\Big[-9450ay^4 + 9450ay^2 - 1152a - 32\Big]\\
&+ x^5(xz - y^2 + 2)^5\Big[-2520y^2 + 574\Big]\\
&+ x^5(xz - y^2 + 2)^6\Big[42a - 42\Big]\\
&+ x^6\Big[-70y^9 + y^7\left(980a + 840\right) + y^5\left(-9390a - 5960\right)\\
&\quad + y^3\left(26352a + 18598\right) + y\left(-29700a - 42708\right)\Big]\\
&+ x^6(xz - y^2 + 2)\Big[y^7\left(840a + 280\right) + y^5\left(22740 - 12660a\right)\\
&\quad + y^3\left(55912a - 101220\right) + y\left(85240 - 52272a\right)\Big]\\
&+ x^6(xz - y^2 + 2)^2\Big[y^7\left(2520a - 2520\right) + y^5\left(26460 - 32220a\right)\\
&\quad + y^3\left(41500a - 37200\right) + y\left(17646 - 15924a\right)\Big]\\
&+ x^6(xz - y^2 + 2)^3\Big[1680ay^7 + 7840ay^5 + y^3\left(-46600a - 13800\right)\\
&\quad + y\left(30200a + 11604\right)\Big]\\
&+ x^6(xz - y^2 + 2)^4\Big[8820y^5 - 5600y^3 + y\left(2184a - 6334\right)\Big]\\
&+ x^6(xz - y^2 + 2)^5\Big[y^3\left(7560 - 7560a\right) + y\left(3780a - 3780\right)\Big]\\
&+ x^6(xz - y^2 + 2)^6\Big[-840ay\Big]\\
&+ x^7\Big[y^8\left(140 - 140a\right) + y^6\left(4090a - 2730\right)\\
&\quad + y^4\left(29360 - 31050a\right) + y^2\left(69544a - 62488\right) + 19252a - 33824\Big]\\
&+ x^7(xz - y^2 + 2)\Big[-560ay^8 + y^6\left(13020a - 6860\right) + y^4\left(45550 - 81190a\right)\\
&\quad + y^2\left(143380a - 75288\right) - 72656a - 29088\Big]\\
&+ x^7(xz - y^2 + 2)^2\Big[y^6\left(3920a - 1960\right) + y^4\left(25200a + 29400\right)\\
&\quad + y^2\left(-67434a - 83016\right) + 16448a + 25328\Big]\\
&+ x^7(xz - y^2 + 2)^3\Big[-5880y^6 + y^4\left(55700 - 11600a\right) + y^2\left(7400a - 44600\right)\\
&\quad - 184a - 6034\Big]\\
&+ x^7(xz - y^2 + 2)^4\Big[y^6\left(2940a - 2940\right) + y^4\left(9800a - 9800\right)\\
&\quad + y^2\left(34950 - 45850a\right) + 10618a - 7550\Big]
\end{aligned}
$$

$$
\begin{aligned}
&+ x^7(xz-y^2+2)^5\left[8820ay^4-3360ay^2-830a+440\right]\\
&+ x^7(xz-y^2+2)^6\left[3780y^2-630\right]\\
&+ x^7(xz-y^2+2)^7\left[-120a-120\right]\\
&+ x^8\Big[y^7\left(-1260a-140\right)+y^5\left(18774a-3454\right)\\
&\qquad + y^3\left(17932-97842a\right)+y\left(182792a-5992\right)\Big]\\
&+ x^8(xz-y^2+2)\Big[840y^7+y^5\left(7260a-19440\right)+y^3\left(108880-3660a\right)\\
&\qquad + y\left(6696a-115536\right)\Big]\\
&+ x^8(xz-y^2+2)^2\Big[y^7\left(1960-1960a\right)+y^5\left(19320a-39900\right)\\
&\qquad + y^3\left(171260-83040a\right)+y\left(80414a-151910\right)\Big]\\
&+ x^8(xz-y^2+2)^3\Big[y^5\left(5880a-7840\right)+y^3\left(69800a-33600\right)\\
&\qquad + y\left(44620-99380a\right)\Big]\\
&+ x^8(xz-y^2+2)^4\Big[-8820ay^5+y^3\left(44100a+13600\right)+y\left(-18600a-5250\right)\Big]\\
&+ x^8(xz-y^2+2)^5\Big[-3528y^5-7840y^3+y\left(18560-4580a\right)\Big]\\
&+ x^8(xz-y^2+2)^6\Big[y^3\left(5880a-5880\right)+y\left(1120-1120a\right)\Big]\\
&+ x^8(xz-y^2+2)^7\Big[1080ay\Big]\\
&+ x^9\Big[y^8\left(140a-140\right)+y^6\left(3920-2240a\right)\\
&\qquad + y^4\left(22830a-37710\right)+y^2\left(91720-61032a\right)-3984a-109376\Big]\\
&+ x^9(xz-y^2+2)\Big[y^6\left(8120-9380a\right)+y^4\left(82200a-95270\right)\\
&\qquad + y^2\left(299520-260790a\right)+164944a-174240\Big]\\
&+ x^9(xz-y^2+2)^2\Big[1960ay^6+y^4\left(-19950a-15510\right)+y^2\left(139500a-90\right)\\
&\qquad - 46340a-504\Big]\\
&+ x^9(xz-y^2+2)^3\Big[3920y^6+y^4\left(-34300a-33600\right)+y^2\left(85260a+94880\right)\\
&\qquad - 22560a+30920\Big]\\
&+ x^9(xz-y^2+2)^4\Big[-9800y^4+y^2\left(24900a-50100\right)-13540a-24615\Big]\\
&+ x^9(xz-y^2+2)^5\Big[y^4\left(8820-8820a\right)+y^2\left(35820a-26460\right)-5080a-3720\Big]\\
&+ x^9(xz-y^2+2)^6\Big[-2940ay^4-3920ay^2+2330a+800\Big]\\
&+ x^9(xz-y^2+2)^7\Big[-2520y^2+160\Big]\\
&+ x^9(xz-y^2+2)^8\Big[135a-135\Big]\\
&+ x^{10}\Big[140ay^7+y^5\left(9470-15050a\right)+y^3\left(127960a-87110\right)\\
&\qquad + y\left(217496-362280a\right)\Big]\\
&+ x^{10}(xz-y^2+2)\Big[-840y^7+y^5\left(7000a+10080\right)+y^3\left(-38420a-68640\right)\\
&\qquad + y\left(27440a+70600\right)\Big]
\end{aligned}
$$

$$
\begin{aligned}
&+ x^{10}(xz - y^2 + 2)^2 \Big[y^5 (26460 - 4200a) + y^3 (-19060a - 174280) \\
&\qquad + y (25680a + 279960) \Big] \\
&+ x^{10}(xz - y^2 + 2)^3 \Big[y^5 (1960a - 1960) + y^3 (12600 - 29240a) \\
&\qquad + y (76380a - 80780) \Big] \\
&+ x^{10}(xz - y^2 + 2)^4 \Big[4900ay^5 + y^3 (34300 - 69300a) + y (94510a - 41110) \Big] \\
&+ x^{10}(xz - y^2 + 2)^5 \Big[y^3 (-7840a - 1960) + y (-10080a - 9060) \Big] \\
&+ x^{10}(xz - y^2 + 2)^6 \Big[5880y^3 + y (3520a - 12340) \Big] \\
&+ x^{10}(xz - y^2 + 2)^7 \Big[y^3 (1680 - 1680a) + y (1120 - 1120a) \Big] \\
&+ x^{10}(xz - y^2 + 2)^8 \Big[-630ay \Big] \\
&+ x^{11} \Big[y^6 (560a - 2940) + y^4 (30900 - 1640a) \\
&\qquad + y^2 (27660a - 96460) - 128224a + 112576 \Big] \\
&+ x^{11}(xz - y^2 + 2) \Big[y^6 (1260a - 1260) + y^4 (76020 - 33420a) \\
&\qquad + y^2 (142320a - 390240) - 104520a - 301480 \Big] \\
&+ x^{11}(xz - y^2 + 2)^2 \Big[-2100ay^6 + y^4 (26600a - 9800) + y^2 (26760 - 93450a) \\
&\qquad + 7320a + 3200 \Big] \\
&+ x^{11}(xz - y^2 + 2)^3 \Big[y^4 (33600a + 7000) + y^2 (13300 - 196140a) + 103960a - 4880 \Big] \\
&+ x^{11}(xz - y^2 + 2)^4 \Big[y^2 (8010 - 9060a) - 2340a + 14800 \Big] \\
&+ x^{11}(xz - y^2 + 2)^5 \Big[y^4 (3920a - 3920) + y^2 (42420 - 21840a) + 11880a - 19790 \Big] \\
&+ x^{11}(xz - y^2 + 2)^6 \Big[y^2 (3920 - 5880a) - 3040a - 1680 \Big] \\
&+ x^{11}(xz - y^2 + 2)^7 \Big[2520ay^2 - 1260a + 560 \Big] \\
&+ x^{11}(xz - y^2 + 2)^8 \Big[630y^2 + 140 \Big] \\
&+ x^{11}(xz - y^2 + 2)^9 \Big[-70a - 70 \Big] \\
&+ x^{12} \Big[140ay^7 + y^5 (3220a - 3500) + y^3 (96360 - 77320a) \\
&\qquad + y (339080a - 402240) \Big] \\
&+ x^{12}(xz - y^2 + 2) \Big[y^5 (-10080a - 2520) + y^3 (68320a + 8620) \\
&\qquad + y (7440a - 55760) \Big] \\
&+ x^{12}(xz - y^2 + 2)^2 \Big[-4200y^5 + y^3 (77020a + 71940) + y (-238080a - 149880) \Big] \\
&+ x^{12}(xz - y^2 + 2)^3 \Big[y^5 (2800 - 2800a) + y^3 (11200a - 14000) \\
&\qquad + y (16560 - 18800a) \Big] \\
&+ x^{12}(xz - y^2 + 2)^4 \Big[y^3 (36400a - 30100) + y (99470 - 95040a) \Big] \\
&+ x^{12}(xz - y^2 + 2)^5 \Big[1960ay^3 + y (2100 - 4620a) \Big] \\
&+ x^{12}(xz - y^2 + 2)^6 \Big[-1960y^3 + y (6860a + 7560) \Big]
\end{aligned}
$$

$$
\begin{aligned}
&+ x^{12}(xz - y^2 + 2)^7 \Big[y\,(1960 - 840a) \Big] \\
&+ x^{12}(xz - y^2 + 2)^8 \Big[y\,(630a - 630) \Big] \\
&+ x^{12}(xz - y^2 + 2)^9 \Big[140ay \Big] \\
&+ x^{13} \Big[560y^6 + y^4\,(-11765a - 13715) + y^2\,(31120a + 61280) \\
&\qquad + 265312a - 20976 \Big] \\
&+ x^{13}(xz - y^2 + 2) \Big[y^6\,(560a - 560) + y^4\,(2800a - 21700) \\
&\qquad + y^2\,(32680a + 240600) - 74400a + 214400 \Big] \\
&+ x^{13}(xz - y^2 + 2)^2 \Big[y^4\,(15960 - 20160a) + y^2\,(47160a - 31920) + 37160a - 69160 \Big] \\
&+ x^{13}(xz - y^2 + 2)^3 \Big[-7000ay^4 + y^2\,(141260a - 70280) - 125680a - 77280 \Big] \\
&+ x^{13}(xz - y^2 + 2)^4 \Big[2100y^4 + 5600ay^2 - 5460a + 2080 \Big] \\
&+ x^{13}(xz - y^2 + 2)^5 \Big[y^2\,(2940a - 18900) + 574a + 19600 \Big] \\
&+ x^{13}(xz - y^2 + 2)^6 \Big[y^2\,(1960a - 1960) - 1400a - 1470 \Big] \\
&+ x^{13}(xz - y^2 + 2)^7 \Big[-560ay^2 + 2100a - 980 \Big] \\
&+ x^{13}(xz - y^2 + 2)^8 \Big[140a + 140 \Big] \\
&+ x^{13}(xz - y^2 + 2)^9 \Big[-70 \Big] \\
&+ x^{13}(xz - y^2 + 2)^{10} \Big[14a - 14 \Big] \\
&+ x^{14} \Big[y^5\,(1470a - 980) + y^3\,(17320a - 43120) \\
&\qquad + y\,(356160 - 140480a) \Big] \\
&+ x^{14}(xz - y^2 + 2) \Big[2520ay^5 + y^3\,(25760 - 58800a) + y\,(47040 - 18000a) \Big] \\
&+ x^{14}(xz - y^2 + 2)^2 \Big[-840y^5 + y^3\,(-37380a - 9800) + y\,(245840a - 25520) \Big] \\
&+ x^{14}(xz - y^2 + 2)^3 \Big[y^3\,(13720 - 3080a) + y\,(11760a + 12320) \Big] \\
&+ x^{14}(xz - y^2 + 2)^4 \Big[y^3\,(6300 - 6300a) + y\,(31920a - 64470) \Big] \\
&+ x^{14}(xz - y^2 + 2)^5 \Big[840ay^3 + y\,(8680a - 5320) \Big] \\
&+ x^{14}(xz - y^2 + 2)^6 \Big[y\,(-4620a - 560) \Big] \\
&+ x^{14}(xz - y^2 + 2)^7 \Big[-840y \Big] \\
&+ x^{14}(xz - y^2 + 2)^8 \Big[y\,(70 - 70a) \Big] \\
&+ x^{15} \Big[70y^6 + y^4\,(7000a + 2100) + y^2\,(-62720a - 5720) \\
&\qquad - 235840a - 127360 \Big] \\
&+ x^{15}(xz - y^2 + 2) \Big[y^4\,(1050a - 1330) + y^2\,(-63840a - 72240) \\
&\qquad + 132160a + 47200 \Big] \\
&+ x^{15}(xz - y^2 + 2)^2 \Big[y^4\,(4200a - 4200) + y^2\,(31080 - 19320a) - 11360a - 79680 \Big] \\
&+ x^{15}(xz - y^2 + 2)^3 \Big[-560ay^4 + y^2\,(33740 - 43820a) + 57680a - 68880 \Big]
\end{aligned}
$$

$$
\begin{aligned}
&+ x^{15}(xz - y^2 + 2)^4 \left[y^2 (1260a + 840) + 16940a - 3500 \right] \\
&+ x^{15}(xz - y^2 + 2)^5 \left[2940y^2 - 6160a + 5040 \right] \\
&+ x^{15}(xz - y^2 + 2)^6 \left[y^2 (140a - 140) + 1120a - 2520 \right] \\
&+ x^{15}(xz - y^2 + 2)^7 \left[-560a - 560 \right] \\
&+ x^{15}(xz - y^2 + 2)^8 \left[-140a \right] \\
&+ x^{16} \Big[y^5 (490 - 490a) + y^3 (2240a + 3360) \\
&\qquad + y (-6560a - 172640) \Big] \\
&+ x^{16}(xz - y^2 + 2) \left[140ay^5 + y^3 (21840a - 16240) + y (4480a - 17920) \right] \\
&+ x^{16}(xz - y^2 + 2)^2 \left[y^3 (4060a - 560) + y (52080 - 112560a) \right] \\
&+ x^{16}(xz - y^2 + 2)^3 \left[-3080y^3 + y (-7280a - 3360) \right] \\
&+ x^{16}(xz - y^2 + 2)^4 \left[y^3 (140 - 140a) + y (16800 - 3360a) \right] \\
&+ x^{16}(xz - y^2 + 2)^5 \left[y (1680 - 1680a) \right] \\
&+ x^{16}(xz - y^2 + 2)^6 \left[560ay \right] \\
&+ x^{17} \left[y^4 (140 - 840a) + y^2 (40320a - 15680) + 107600a - 103040 \right] \\
&+ x^{17}(xz - y^2 + 2) \left[1050y^4 + y^2 (23520a + 8400) - 67200a - 15680 \right] \\
&+ x^{17}(xz - y^2 + 2)^2 \left[y^4 (70a - 70) + y^2 (3360a - 12600) - 13440a + 32480 \right] \\
&+ x^{17}(xz - y^2 + 2)^3 \left[y^2 (5040a - 5040) - 8960a - 21840 \right] \\
&+ x^{17}(xz - y^2 + 2)^4 \left[-840ay^2 - 8400a - 5040 \right] \\
&+ x^{17}(xz - y^2 + 2)^5 \left[1680a \right] \\
&+ x^{17}(xz - y^2 + 2)^6 \left[560 \right] \\
&+ x^{18} \Big[y^5 (14 - 14a) + y^3 (2800 - 1120a) \\
&\qquad + y (29120a + 43680) \Big] \\
&+ x^{18}(xz - y^2 + 2) \left[y^3 (2800 - 2800a) + 2240y \right] \\
&+ x^{18}(xz - y^2 + 2)^2 \left[560ay^3 + y (25200a - 18480) \right] \\
&+ x^{18}(xz - y^2 + 2)^3 \left[1120ay \right] \\
&+ x^{18}(xz - y^2 + 2)^4 \left[-1680y \right] \\
&+ x^{19} \left[-140ay^4 + y^2 (7840 - 11200a) - 24640a - 38080 \right] \\
&+ x^{19}(xz - y^2 + 2) \left[-2800ay^2 + 14560a - 8960 \right] \\
&+ x^{19}(xz - y^2 + 2)^2 \left[1680y^2 + 7840a + 4480 \right] \\
&+ x^{19}(xz - y^2 + 2)^3 \left[-2240 \right] \\
&+ x^{19}(xz - y^2 + 2)^4 \left[1120a - 1120 \right] \\
&+ x^{20} \left[-560y^3 + y (-10080a - 4480) \right]
\end{aligned}
$$

$$
\begin{aligned}
&+ x^{20}(xz - y^2 + 2)^2\Big[y\,(2240 - 2240a)\Big] \\
&+ x^{21}\Big[y^2\,(1120a - 1120) + 2240a - 6720\Big] \\
&+ x^{21}(xz - y^2 + 2)\Big[-1120a - 1120\Big] \\
&+ x^{21}(xz - y^2 + 2)^2\Big[-1120a\Big] \\
&+ x^{22}\Big[1120ay\Big] \\
&+ x^{23}\Big[448\Big] \\
&+ \Big[xy^2\,(126a - 126) + x\,(49 - 49a) + y^5\,(252a - 252) \\
&\qquad + y^3\,(700 - 700a) + y\,(294a - 322)\Big] \\
&\qquad \times \Big[(1 - a)\big(ax(xz - y^2 + 2) + y\big)^2 \\
&\qquad + \big((a - 1)x^2 - a\big)(xz - y^2 + 2) \\
&\qquad + ax(1 - x^2)y - 2x^4 + (6 - 2a)x^2 - 2(1 - a)\Big] \\
&+ \Big[x^2y\,(126 - 126a) + xy^4\,(630 - 630a) \\
&\qquad + xy^2\,(1050a - 1050) + x\,(161 - 147a)\Big] \\
&\qquad \times \Big[(1 - a)\big(ax(xz - y^2 + 2) + y\big)^2 \\
&\qquad + \big((a - 1)x^2 - a\big)(xz - y^2 + 2) \\
&\qquad + ax(1 - x^2)y - 2x^4 + (6 - 2a)x^2 - 2(1 - a)\Big]^2 \\
&+ \Big[x^3\,(42a - 42) + x^2y^3\,(840a - 840) \\
&\qquad + x^2y\,(700 - 700a)\Big] \\
&\qquad \times \Big[(1 - a)\big(ax(xz - y^2 + 2) + y\big)^2 \\
&\qquad + \big((a - 1)x^2 - a\big)(xz - y^2 + 2) \\
&\qquad + ax(1 - x^2)y - 2x^4 + (6 - 2a)x^2 - 2(1 - a)\Big]^3 \\
&+ \Big[x^3y^2\,(630 - 630a) + x^3\,(175a - 175)\Big] \\
&\qquad \times \Big[(1 - a)\big(ax(xz - y^2 + 2) + y\big)^2 \\
&\qquad + \big((a - 1)x^2 - a\big)(xz - y^2 + 2) \\
&\qquad + ax(1 - x^2)y - 2x^4 + (6 - 2a)x^2 - 2(1 - a)\Big]^4 \\
&+ \Big[x^4y\,(252a - 252)\Big] \\
&\qquad \times \Big[(1 - a)\big(ax(xz - y^2 + 2) + y\big)^2 \\
&\qquad + \big((a - 1)x^2 - a\big)(xz - y^2 + 2) \\
&\qquad + ax(1 - x^2)y - 2x^4 + (6 - 2a)x^2 - 2(1 - a)\Big]^5 \\
&+ \Big[x^5\,(42 - 42a)\Big] \\
&\qquad \times \Big[(1 - a)\big(ax(xz - y^2 + 2) + y\big)^2 \\
&\qquad + \big((a - 1)x^2 - a\big)(xz - y^2 + 2) \\
&\qquad + ax(1 - x^2)y - 2x^4 + (6 - 2a)x^2 - 2(1 - a)\Big]^6
\end{aligned}
$$

Inverse map

The variables (X, Y, Z) are independent input coordinates of the inverse.

Component 1.

$$(\Phi_H^{-1})_1(X,Y,Z) = X^3 - 3X + Y.$$

Component 2.

$$\begin{aligned}
(\Phi_H^{-1})_2(X,Y,Z) = & -X \\
& + (X^3 - 3X + Y)\Big[X^2 - 2\Big] \\
& + (X^3 - 3X + Y)^2\Big[X^7(132a - 132) + X^5(630 - 630a) + X^3(742a - 788) \\
& \quad + X(181 - 120a)\Big] \\
& + (X^3 - 3X + Y)^3\Big[-165X^8a + 1050X^6a + X^4(68 - 2058a) + X^2(1171a - 196) \\
& \quad - 40a - 44\Big] \\
& + (X^3 - 3X + Y)^4\Big[308X^9 - 2040X^7 + X^5(162a + 4078) + X^3(-522a - 2190) \\
& \quad + X(196a - 20)\Big] \\
& + (X^3 - 3X + Y)^5\Big[X^{10}(616a - 616) + X^8(4410 - 4410a) \\
& \quad + X^6(9892a - 10340) + X^4(8510 - 6810a) \\
& \quad + X^2(495a - 1644) + 32a\Big] \\
& + (X^3 - 3X + Y)^6\Big[-1176X^{11}a + 9100X^9a + X^7(812 - 24444a) + X^5(26474a - 3374) \\
& \quad + X^3(3082 - 9998a) + X(384a - 464)\Big] \\
& + (X^3 - 3X + Y)^7\Big[1764X^{12} - 14700X^{10} + X^8(1176a + 42973) \\
& \quad + X^6(-5572a - 51450) + X^4(7112a + 21413) \\
& \quad + X^2(224 - 2576a) + 64a - 64\Big] \\
& + \big(Z + (1-a)(42X^3 - 49X)\big)\Bigg\{ \\
& (X^3 - 3X + Y)^2\Big[a\Big] \\
& + (X^3 - 3X + Y)^3\Big[2X\Big] \\
& + (X^3 - 3X + Y)^4\Big[X^2(2a - 2) - 5a - 5\Big] \\
& + (X^3 - 3X + Y)^5\Big[-4X^3a + 10Xa\Big] \\
& + (X^3 - 3X + Y)^6\Big[10X^4 - 30X^2 + 4a\Big] \\
& + (X^3 - 3X + Y)^7\Big[X^5(28a - 28) + X^3(100 - 100a) + X(48a - 58)\Big] \\
& + (X^3 - 3X + Y)^8\Big[-84X^6a + 350X^4a + X^2(28 - 322a) + 16a - 16\Big]\Bigg\} \\
& + \big(Z + (1-a)(42X^3 - 49X)\big)^2\Bigg\{ \\
& (X^3 - 3X + Y)^9\Big[a - 1\Big]\Bigg\}
\end{aligned}$$

Component 3.

$$
\begin{aligned}
(\Phi_H^{-1})_3(X,Y,Z) = {} & -132X^7a + 630X^5a + X^3(46 - 788a) + X(181a - 62) \\
& + (X^3 - 3X + Y)\Big[X^8(429 - 264a) + X^6(1260a - 2310) \\
& \quad + X^4(3562 - 1411a) + X^2(29a - 1326) + 44a - 2\Big] \\
& + (X^3 - 3X + Y)^2\Big[X^9(902a - 572) + X^7(3828 - 5928a) \\
& \quad + X^5(12184a - 8472) + X^3(6618 - 7690a) + X(511a - 983)\Big] \\
& + (X^3 - 3X + Y)^3\Big[-17424X^{14}a + 166320X^{12}a + X^{10}(11528 - 605878a) \\
& \quad + X^8(1047834a - 69984) + X^6(139710 - 865868a) \\
& \quad + X^4(299772a - 105335) + X^2(20569 - 35826a) + 152a - 208\Big] \\
& + (X^3 - 3X + Y)^4\Big[43560X^{15} - 485100X^{13} + X^{11}(31900a + 2096244) \\
& \quad + X^9(-245334a - 4385832) + X^7(658276a + 4486976) \\
& \quad + X^5(-712878a - 1953944) + X^3(263450a + 219028) \\
& \quad + X(6488 - 16752a)\Big] \\
& + (X^3 - 3X + Y)^5\Big[X^{16}(108537a - 108537) + X^{14}(1273140 - 1273140a) \\
& \quad + X^{12}(5868612a - 5959804) + X^{10}(14119118 - 13362654a) \\
& \quad + X^8(15411509a - 17656533) + X^6(10962294 - 8138722a) \\
& \quad + X^4(1374148a - 2740902) + X^2(51300a + 126088) - 2112a - 336\Big] \\
& + (X^3 - 3X + Y)^6\Big[-264264X^{17}a + 3260400X^{15}a + X^{13}(266764 - 16210456a) \\
& \quad + X^{11}(41388032a - 2375388) + X^9(7724210 - 57080708a) \\
& \quad + X^7(40668068a - 11053496) + X^5(6576582 - 12668156a) \\
& \quad + X^3(1004302a - 1223192) + X(5408a + 23584)\Big] \\
& + (X^3 - 3X + Y)^7\Big[608608X^{18} - 7889700X^{16} + X^{14}(653968a + 41104248) \\
& \quad + X^{12}(-6220072a - 109458596) + X^{10}(22150686a + 156150628) \\
& \quad + X^8(-36238916a - 113101396) + X^6(26789748a + 33798166) \\
& \quad + X^4(-7506802a - 1094458) + X^2(397096a - 294864) + 2816\Big] \\
& + (X^3 - 3X + Y)^8\Big[X^{19}(1233232a - 1233232) + X^{17}(16805880 - 16805880a) \\
& \quad + X^{15}(93184056a - 94560256) + X^{13}(282933272 - 268948700a) \\
& \quad + X^{11}(428059452a - 482380668) + X^9(464533414 - 364272396a) \\
& \quad + X^7(143908164a - 232745724) + X^5(48510018 - 14794632a) \\
& \quad + X^3(-1777104a - 1872200) + X(74624a - 30464)\Big] \\
& + (X^3 - 3X + Y)^9\Big[-1685992X^{20}a + 24392200X^{18}a + X^{16}(2061136 - 147044758a) \\
& \quad + X^{14}(477435896a - 22612296) + X^{12}(96292392 - 900050894a) \\
& \quad + X^{10}(987073522a - 199641004) + X^8(207476496 - 592378812a) \\
& \quad + X^6(163578942a - 100691100) + X^4(17775911 - 11329319a) \\
& \quad + X^2(-183872a - 474816) + 6656a - 1536\Big] \\
& + (X^3 - 3X + Y)^{10}\Big[2535456X^{21} - 37835840X^{19} + X^{17}(3350032a + 233477160) \\
& \quad + X^{15}(-38305456a - 767587520) + X^{13}(172213356a + 1441705300) \\
& \quad + X^{11}(-384911884a - 1534382556) + X^9(446541820a + 846197492)
\end{aligned}
$$

$$
\begin{aligned}
&+ X^7\left(-256277268a - 174468888\right) + X^5\left(60287936a - 12454740\right)\\
&+ X^3\left(3133312 - 2883680a\right) + X\left(-7680a - 47104\right)\Big]\\
&+ \left(X^3 - 3X + Y\right)^{11}\Big[X^{22}\left(3556224a - 3556224\right) + X^{20}\left(55072080 - 55072080a\right)\\
&+ X^{18}\left(355888176a - 360827376\right) + X^{16}\left(1300812212 - 1241692340a\right)\\
&+ X^{14}\left(2525752824a - 2808210384\right) + X^{12}\left(3696110096 - 3009217008a\right)\\
&+ X^{10}\left(1978554914a - 2878553048\right) + X^8\left(1208850804 - 589652784a\right)\\
&+ X^6\left(19578546a - 215967752\right) + X^4\left(13022944a + 5959328\right)\\
&+ X^2\left(379776 - 569856a\right) - 4096\Big]\\
&+ \left(X^3 - 3X + Y\right)^{12}\Big[-4148928X^{23}a + 66679200X^{21}a + X^{19}\left(5630688 - 457616880a\right)\\
&+ X^{17}\left(1749681416a - 70284816\right) + X^{15}\left(355487160 - 4074946232a\right)\\
&+ X^{13}\left(5928291068a - 935952332\right) + X^{11}\left(1373753220 - 5294231852a\right)\\
&+ X^9\left(2711832284a - 1117135236\right) + X^7\left(464133572 - 672216524a\right)\\
&+ X^5\left(46027520a - 78419040\right) + X^3\left(978688a + 2655744\right)\\
&+ X\left(10240 - 59392a\right)\Big]\\
&+ \left(X^3 - 3X + Y\right)^{13}\Big[3111696X^{24} - 51861600X^{22} + X^{20}\left(4148928a + 367698744\right)\\
&+ X^{18}\left(-54232416a - 1444921800\right) + X^{16}\left(291363408a + 3433470817\right)\\
&+ X^{14}\left(-831187784a - 5037568284\right) + X^{12}\left(1358704928a + 4432878198\right)\\
&+ X^{10}\left(-1278517800a - 2096949316\right) + X^8\left(651940688a + 350528409\right)\\
&+ X^6\left(53532864 - 150360224a\right) + X^4\left(8222592a - 10236800\right)\\
&+ X^2\left(28672a + 301056\right) - 4096a\Big]\\
&+ \Big(Z + (1-a)(42X^3 - 49X)\Big)\Bigg\{\\
&- 1\\
&+ \left(X^3 - 3X + Y\right)\Big[-2X\Big]\\
&+ \left(X^3 - 3X + Y\right)^2\Big[-4X^2 + a\Big]\\
&+ \left(X^3 - 3X + Y\right)^3\Big[-264X^7 + 1260X^5 + X^3\left(-96a - 1472\right) + X\left(132a + 212\right)\Big]\\
&+ \left(X^3 - 3X + Y\right)^4\Big[X^8\left(198a - 198\right) + X^6\left(420 - 420a\right) + X^4\left(950 - 990a\right)\\
&+ X^2\left(1402a - 1570\right) + 27a + 56\Big]\\
&+ \left(X^3 - 3X + Y\right)^5\Big[-572X^9a + 3960X^7a + X^5\left(112 - 9240a\right) + X^3\left(8000a - 384\right)\\
&+ X\left(404 - 1724a\right)\Big]\\
&+ \left(X^3 - 3X + Y\right)^6\Big[1716X^{10} - 12870X^8 + X^6\left(336a + 33684\right) + X^4\left(-1360a - 35390\right)\\
&+ X^2\left(1782a + 11594\right) - 408a + 24\Big]\\
&+ \left(X^3 - 3X + Y\right)^7\Big[X^{11}\left(5304a - 5304\right) + X^9\left(42900 - 42900a\right)\\
&+ X^7\left(118800a - 121968\right) + X^5\left(141120 - 126112a\right)\\
&+ X^3\left(37408a - 55512\right) + X\left(512a + 3016\right)\Big]\\
&+ \left(X^3 - 3X + Y\right)^8\Big[-16796X^{12}a + 145860X^{10}a + X^8\left(11880 - 460746a\right)\\
&+ X^6\left(632940a - 62664\right) + X^4\left(93244 - 345978a\right)
\end{aligned}
$$

$$
\begin{aligned}
&\quad + X^2\,(48088a - 33870) - 32a - 576\Big] \\
&+ (X^3 - 3X + Y)^9\Big[54264X^{13} - 503880X^{11} + X^9\,(39468a + 1709268) \\
&\quad + X^7\,(-230472a - 2538720) + X^5\,(410256a + 1516356) \\
&\quad + X^3\,(-221928a - 226764) + X\,(17760a - 6480)\Big] \\
&+ (X^3 - 3X + Y)^{10}\Big[X^{14}\,(73752a - 73752) + X^{12}\,(765660 - 765660a) \\
&\quad + X^{10}\,(2984664a - 3045648) + X^8\,(5751450 - 5330970a) \\
&\quad + X^6\,(4205664a - 5138820) + X^4\,(1824990 - 1119522a) \\
&\quad + X^2\,(-32136a - 106152) + 2304a - 384\Big] \\
&+ (X^3 - 3X + Y)^{11}\Big[-123872X^{15}a + 1333920X^{13}a + X^{11}\,(116928 - 5562648a) \\
&\quad + X^9\,(11211428a - 855288) + X^7\,(2078424 - 11045376a) \\
&\quad + X^5\,(4634628a - 1865520) + X^3\,(521328 - 465580a) \\
&\quad + X\,(-1920a - 13056)\Big] \\
&+ (X^3 - 3X + Y)^{12}\Big[204624X^{16} - 2316720X^{14} + X^{12}\,(216384a + 10108812) \\
&\quad + X^{10}\,(-1698816a - 21162540) + X^8\,(4610592a + 21407148) \\
&\quad + X^6\,(-5046900a - 9000432) + X^4\,(2035068a + 664596) \\
&\quad + X^2\,(98784 - 146304a) - 1536\Big] \\
&+ (X^3 - 3X + Y)^{13}\Big[X^{17}\,(296352a - 296352) + X^{15}\,(3528000 - 3528000a) \\
&\quad + X^{13}\,(16547496a - 16850904) + X^{11}\,(41184696 - 38661000a) \\
&\quad + X^9\,(46773048a - 54298692) + X^7\,(37019472 - 27293448a) \\
&\quad + X^5\,(5595072a - 10911180) + X^3\,(257472a + 663936) \\
&\quad + X\,(3840 - 22272a)\Big] \\
&+ (X^3 - 3X + Y)^{14}\Big[-296352X^{18}a + 3704400X^{16}a + X^{14}\,(296352 - 18843048a) \\
&\quad + X^{12}\,(50007244a - 2638944) + X^{10}\,(8729448 - 73544100a) \\
&\quad + X^8\,(58147908a - 13293504) + X^6\,(9417912 - 21264628a) \\
&\quad + X^4\,(2055648a - 2604000) + X^2\,(10752a + 112896) - 2048a\Big]\Bigg\} \\
&+ \big(Z + (1 - a)(42X^3 - 49X)\big)^2\Bigg\{ \\
&(X^3 - 3X + Y)^3\Big[a - 1\Big] \\
&+ (X^3 - 3X + Y)^4\Big[4Xa\Big] \\
&+ (X^3 - 3X + Y)^5\Big[10\Big] \\
&+ (X^3 - 3X + Y)^7\Big[-18a + 8\Big] \\
&+ (X^3 - 3X + Y)^8\Big[X\,(6 - 6a)\Big] \\
&+ (X^3 - 3X + Y)^9\Big[X^2\,(6a - 6) - 4a - 12\Big] \\
&+ (X^3 - 3X + Y)^{10}\Big[-792X^7a + 3780X^5a + X^3\,(276 - 4728a) + X\,(1086a - 366)\Big] \\
&+ (X^3 - 3X + Y)^{11}\Big[990X^8 - 6300X^6 + X^4\,(408a + 11940) + X^2\,(-1176a - 5850) \\
&\quad + 264a - 24\Big]
\end{aligned}
$$

$$
\begin{aligned}
&+ (X^3 - 3X + Y)^{12} \Big[X^9 (1848a - 1848) + X^7 (12240 - 12240a) \\
&\quad + X^5 (24468a - 25440) + X^3 (16272 - 13140a) \\
&\quad + X (-120a - 1056) \Big] \\
&+ (X^3 - 3X + Y)^{13} \Big[-3696X^{10}a + 26460X^8 a + X^6 (2688 - 62040a) \\
&\quad + X^4 (51060a - 10200) + X^2 (6894 - 9864a) - 192 \Big] \\
&+ (X^3 - 3X + Y)^{14} \Big[7056X^{11} - 54600X^9 + X^7 (4872a + 141792) \\
&\quad + X^5 (-20244a - 138600) + X^3 (18492a + 41496) \\
&\quad + X (480 - 2784a) \Big] \\
&+ (X^3 - 3X + Y)^{15} \Big[X^{12} (10584a - 10584) + X^{10} (88200 - 88200a) \\
&\quad + X^8 (257838a - 264894) + X^6 (342132 - 308700a) \\
&\quad + X^4 (128478a - 171150) + X^2 (1344a + 14112) - 384a \Big] \Big\} \\
&+ \big(Z + (1 - a)(42X^3 - 49X) \big)^3 \Big\{ \\
&(X^3 - 3X + Y)^{10} \Big[-2 \Big] \\
&+ (X^3 - 3X + Y)^{11} \Big[X (4a - 4) \Big] \\
&+ (X^3 - 3X + Y)^{12} \Big[-4X^2 a + 10a \Big] \\
&+ (X^3 - 3X + Y)^{13} \Big[8X^3 - 20X \Big] \\
&+ (X^3 - 3X + Y)^{14} \Big[X^4 (20a - 20) + X^2 (60 - 60a) - 8 \Big] \\
&+ (X^3 - 3X + Y)^{15} \Big[-56X^5 a + 200X^3 a + X (20 - 116a) \Big] \\
&+ (X^3 - 3X + Y)^{16} \Big[168X^6 - 700X^4 + X^2 (56a + 588) - 32a \Big] \Big\} \\
&+ \big(Z + (1 - a)(42X^3 - 49X) \big)^4 \Big\{ \\
&(X^3 - 3X + Y)^{17} \Big[-a \Big] \Big\}
\end{aligned}
$$

3.5 Further transfers from the same base coordinate

For every $N \geq 1$ and $p \in K[x, y]$, the coordinate

$$H(x, p(x, y) + x^N z, y)$$

satisfies Theorem 1.1: modulo the nilradical of $K[x]/(x^N)$ its centre is $-y$. The finite algorithm constructs a companion and a polynomial inverse, retaining all powers of x up to x^N. This is a sufficient construction for the specified inputs; it does not assert that every embedding of $\mathbb{A}^1$ into $\mathbb{A}^3$ admits such a presentation.

3.6 Computational certificate

The file `verify.py` performs exact arithmetic in $\mathbb{Q}[a, x, y, z, t]/(a^2 - a + 1)$ using SymPy. It checks:

1. the exact divisions in (12) and both harmonic directions;
2. the five identities (14)–(18), the outer substitutions, and the transfer identities;
3. all components of $\Phi_T(\gamma_T(t)) = (t, 0, 0)$ and $\Phi_H(\gamma_H(t)) = (t, 0, 0)$, and both inverse substitutions on the line $(t, 0, 0)$;
4. the component degrees and the relative Jacobian determinants.

The polynomial certificates, cancellation argument, and factorization (27) establish the full inverse identities. The successful run is recorded in `verification.log`. To reproduce it:

```
python -m pip install -r requirements.txt
python -u verify.py
```

The displayed automorphisms are defined over $K \subset \mathbb{C}$; their coefficients do not define a rectification by real polynomial automorphisms.

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